Example 5.16

Find the equation of the parabola with focus $\left(-\sqrt{2},0\right)$ and directrix $x=\sqrt{2}$.

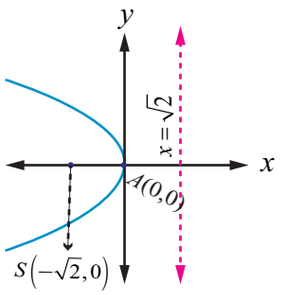


Fig.5.32

Solution

Parabola is open left and axis of symmetry as x-axis and vertex $(0,0)$.

Then the equation of the required parabola is

$$(y-0)^2 = -4\sqrt{2}(x-0)$$

$$\Rightarrow \quad y^2 = -4\sqrt{2}\,x.$$

Example 5.17

Find the equation of the parabola whose vertex is $(5,-2)$ and focus $(2,-2)$.

Solution

Given vertex $A(5,-2)$ and focus $S(2,-2)$ and the focal distance $AS = a = 3$.

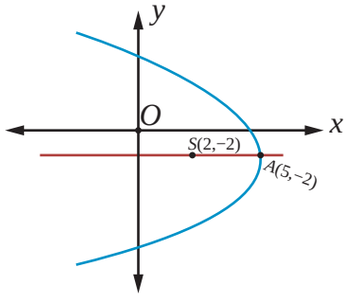


Fig.5.33

Parabola is open left and symmetric about the line parallel to x-axis.

Then, the equation of the required parabola is

$$(y+2)^2 = -4(3)(x-5)$$

$$\Rightarrow \quad y^2+4y+4 = -12x+60$$

$$\Rightarrow \quad y^2+4y+12x-56 = 0.$$

Example 5.18

Find the equation of the parabola with vertex $(-1,-2)$, axis parallel to y-axis and passing through $(3,6)$.

Solution

Since axis is parallel to y-axis the required equation of the parabola is

$$(x+1)^2 = 4a(y+2).$$

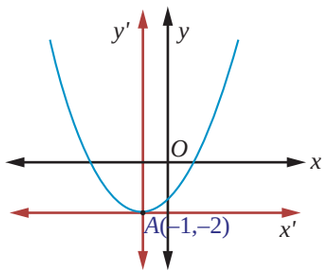


Fig.5.34

Since this passes through (3,6), we get

$$(3+1)^2 = 4a(6+2)$$

$$\Rightarrow \quad a = \frac{1}{2}.$$

Then the equation of parabola is $(x+1)^2 = 2(y+2)$ which on simplifying yields,

$$x^2+2x-2y-3 = 0.$$

Example 5.19

Find the vertex, focus, directrix, and length of the latus rectum of the parabola $x^2-4x-5y-1=0$.

Solution

For the parabola,

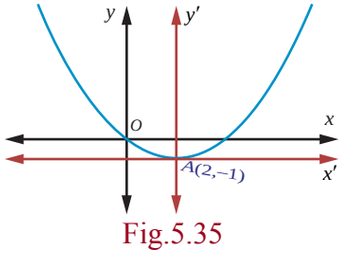


Fig.5.35

$$x^2-4x-5y-1 = 0$$

$$\Rightarrow \quad x^2-4x = 5y+1$$

$$\Rightarrow \quad x^2-4x+4 = 5y+1+4.$$

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